BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 105 · Lesson 105 · RAM, ROM and Virtual Storage: Different Roles

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Where are running program instructions and data commonly held?

Q2. What does ROM or firmware flash retain?

Q3. What is a remote network drive an example of?

Q4. Rising edge D-type flip-flop D0 Q1: Q just after edge?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

RAM — Volatile working memory.

ROM — Non-volatile storage for relatively fixed code.

virtual storage — Remote/pooled physical storage presented logically.

virtual memory — OS memory abstraction using paging.

New learning

Understand — RAM holds currently used programs/data and normally loses content without power; ROM or firmware flash retains startup routines.

Apply — OCR virtual storage means logical storage presented from physically remote or pooled devices, such as a network drive or cloud service.

Evaluate — Do not confuse virtual storage with virtual memory. Virtual memory uses secondary storage to extend the apparent addressable memory of processes; virtual storage is remote-file presentation.

Worked example

A pupil opens a document from a mounted school network drive. Where is the file physically stored?

1. Explain the choice — On one or more remote network storage devices, even if Finder/Explorer shows a drive like local storage.

We Do · Practise with guidance

A laptop runs out of RAM and the OS moves inactive pages to an SSD. Is this OCR 1.1.3(d) virtual storage?

Explain the choice

No. It is virtual memory under 1.2.1, using secondary storage for memory pages.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Explain the role and volatility of RAM and ROM/firmware storage. [6]

Question 2

Define virtual storage and give an example, two benefits and a disadvantage. [8]

Question 3

Correct a pupil's claim that 'virtual storage is extra RAM created when memory fills up'. [6]

Exit Ticket · Answer before checking

OCR H446 1.1.3(d) virtual storage is another term for swapping RAM pages.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. RAM, plus CPU caches/registers.

Q2. Startup/firmware instructions without power.

Q3. Virtual storage.

Q4. 0, because D is sampled at the active edge.

You Do pages · Independent questions

Q1. RAM active programs/data [2], typically volatile [1], ROM/firmware boot instructions [2], non-volatile [1].

Q2. Remote physical storage appears local/logical [2], network drive/cloud [2], access and scalability [2], dependency/network/security concern [2].

Q3. Virtual storage means remote file store [2], virtual memory uses secondary storage for memory pages [2], they are different concepts in OCR spec [2].

Exit Ticket · Model answer

False. That describes virtual memory, not virtual storage.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: